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# LLM-Powered Automation of a Dark Matter Constraint Repository

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## Abstract

1 Dark matter constraint repositories are critical community infrastructure, giving  
2 experimentalists and theorists a shared landscape of existing bounds. Yet the  
3 most widely-used repositories are maintained by individual volunteers, creating  
4 a sustainability risk as the pace of new results accelerates. We present a large  
5 language model (LLM) pipeline that monitors arXiv, extracts limit curves from  
6 papers, integrates them as code, and opens pull requests (PRs) for human review.  
7 On a 346-paper benchmark whose ground truth is the upstream-curated repository  
8 itself, the pipeline classifies the coupling type correctly for 90.5% of papers and  
9 reaches a median coupling residual of 0.33 dex (a factor of two for 48% of curves),  
10 with 76% mean mass-range coverage. Three design choices drive this: (i) treating  
11 each extraction as a noisy sample and reconciling several reads by a consensus  
12 vote with source-quality tiering; (ii) a convention-canonicalization layer whose  
13 per-file conversion factors were derived with a guided physics-derivation assistant  
14 (GPD), code-verified and citation-audited; and (iii) a scoring methodology that  
15 separates genuine extraction error from non-comparability. Strikingly, figure-vision  
16 extraction (0.34 dex) now matches text extraction (0.33 dex): reading log-log  
17 plots is no longer the bottleneck. The remaining difficulty is concentrated in rare  
18 coupling types with idiosyncratic conventions (macro-averaged residual 1.1 dex).  
19 The pipeline is deployed and has generated limit proposals; none have merged —  
20 governance of AI-generated scientific data is itself an unsolved problem.

## 21 1 Introduction

22 The experimental search for ultralight dark matter is undergoing a “renaissance” [1], with new  
23 proposals targeting masses from  $10^{-22}$  eV to several eV using techniques from quantum-limited  
24 cavities to dish antennas and nuclear magnetic resonance [2, 3]. The result is an ever-denser landscape  
25 of constraints across at least 15 distinct *coupling types* (ways a dark matter candidate can interact  
26 with Standard Model particles, e.g. axion-photon or dark photon kinetic mixing) [4], each with its  
27 own conventions for units, parameterisations, and confidence levels.

28 Dark matter constraint repositories compile these bounds into publication-quality *exclusion plots*  
29 — diagrams showing which regions of mass-coupling parameter space have been ruled out — that  
30 appear in a large fraction of papers in the field [5, 4, 6]. The AxionLimits repository [5] tracks over  
31 600 limits and 160 projected sensitivities, yet is maintained by a single volunteer who must identify  
32 new results, extract data from figures or tables, convert between parameterisations (e.g. kinetic mixing  
33  $\chi$  vs. coupling  $g$  [4]), write plotting code, and regenerate figures. With multiple relevant papers  
34 appearing weekly, this concentrates critical community infrastructure on a single point of failure.

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\*This paper was drafted with the assistance of large language models (Claude) under human supervision and editorial control.

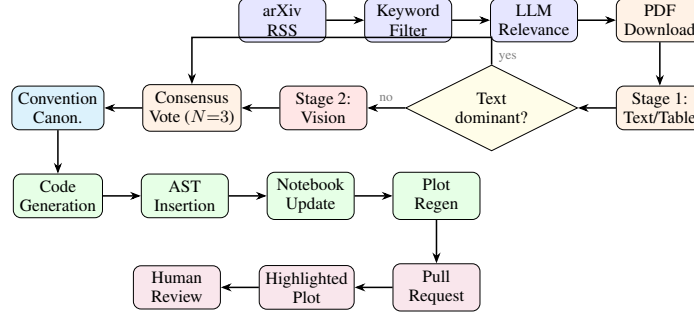


Figure 1: Pipeline architecture. Papers flow from arXiv through keyword and LLM filtering, two-stage extraction (text-first, vision fallback), multi-read consensus voting, physics convention canonicalization, code generation with AST-based insertion, plot regeneration, and finally a pull request with a highlighted constraint plot for human review.

We present a pipeline that partially automates this workflow: (1) **Discovery** via daily arXiv monitoring with keyword filtering and LLM relevance classification; (2) **Extraction** using a two-stage approach (text/tables first, vision fallback for plot-only papers) with multi-read consensus voting and a physics convention-canonicalization layer; (3) **Integration** through automated code generation with Abstract Syntax Tree (AST)-based insertion, notebook updates, and plot regeneration; and (4) **Review** via PRs with highlighted constraint plots. The system also includes a weekly preprint version checker and a historical backfill workflow using INSPIRE-HEP [7].<sup>1</sup>

**Related work.** Chart understanding benchmarks (ChartQA [9], PlotQA [10]) address visual QA but not numerical extraction from dense log–log exclusion plots; document parsers like GROBID [11] handle metadata, not figure digitisation. To our knowledge, no existing system automates the full loop from discovery through extraction, code integration, and human review for a live scientific repository.

**Contributions.** (i) An end-to-end pipeline that turns an arXiv paper into a reviewable code change for a real constraint repository. (ii) An extraction stack that treats LLM reads as noisy samples and reconciles them by consensus with source-quality tiering, and pairs them with a physics convention-canonicalization layer whose conversions were established by a guided physics-derivation assistant rather than by prompting. (iii) A 346-paper benchmark and scoring methodology that grounds truth in the upstream repository and separates extraction error from non-comparability. (iv) The empirical finding that, with these in place, log–log plot reading is no longer the dominant error source — the frontier shifts to deterministic, convention-bound rare coupling types.

## 2 System Architecture

Figure 1 shows the end-to-end pipeline.

### 2.1 Two-Stage Extraction with Consensus Voting

The extraction pipeline prioritises text and table data (Stage 1) over vision-based figure reading (Stage 2), motivated by both cost and accuracy.

**Stage 1: Text and tables.** The PDF is parsed with PyMuPDF [12] and the full text sent to an LLM [8] with a structured prompt requesting JSON output: coupling type, mass–coupling data points, the model’s declared units/convention, and metadata. Prompt-injection defences strip control characters and wrap content in sentinel delimiters.

**Stage 2: Vision fallback.** Vision is invoked only when Stage 1 is not *clearly dominant* (a valid mass window,  $\geq 5$  points, sufficient confidence, non-degenerate). Selected figure pages are rendered to images; the model reads axis labels, tick marks, and curve positions from log–log exclusion plots.

<sup>1</sup>Code, evaluation benchmark, and prompts: <https://anonymous.4open.science/r/AutoAxionLimits-358C>

66 **Consensus across noisy reads.** LLM extraction is non-deterministic, so each paper is read several  
67 times independently ( $N=3$ ) and reconciled. The coupling type is chosen by majority vote; the  
68 reported curve is the *medoid* — the candidate with the smallest median pairwise distance to the others  
69 in log–log space. The medoid is a real sample, not an average, so consensus rejects one-off mis-reads  
70 without ever fabricating a non-physical curve by averaging incompatible ones.

71 **Quality-tier source selection.** When candidates disagree, they are ranked by a lexicographic quality  
72 tuple rather than by raw point count: validity (values in physically allowed ranges), non-degeneracy  
73 (a traced curve must span  $\geq 1$  dex, not a flat artefact), recoverability, *source tier* (table > text >  
74 figure-vision), corroboration between reads, and confidence. A sparse text or table snippet ( $\leq 3$   
75 points, e.g. a headline bound quoted in prose) is demoted below a traceable multi-point figure curve,  
76 so a single quoted number cannot override an actual digitised constraint.

## 77 2.2 Convention Canonicalization

78 The same physics is reported in incompatible  $y$ -variables across papers and even across files within  
79 one coupling type, and the conversion often lives in the repository’s plotting code, not in the paper.  
80 Examples we encountered: axion–nucleon couplings stored as the derivative coupling  $g_{aNN} =$   
81  $C_N/2f_a$  [GeV $^{-1}$ ] but plotted dimensionless after an in-code  $\times 2m_N$  (with a per-file  $\times m_N$  exception  
82 for one detector); scalar/dilaton bounds stored as a genuine Damour–Donoghue  $d_e \sim 10^{-3}$  in some  
83 files but as a fifth-force strength  $\alpha \sim 10^7$  in others (related by  $d_e = Q^{-1}\sqrt{\alpha}$ , a 10–15 dex *apparent*  
84 offset that is not an error); dark-photon kinetic mixing reported as  $\varepsilon^2$  vs.  $\chi = \varepsilon$ ; and large sentinel  
85 values ( $10^{20}, 10^{30}, 10^{99}$ ) that are `fill_between` polygon-closing vertices, not data.

86 These are *deterministic* mismatches: no amount of prompting fixes them, because the conversion  
87 depends on per-file repository storage that the paper never states. We encode them in a per-file  
88 convention registry that the comparator (and the production pipeline) applies to canonicalize both  
89 the extracted and reference curves before any residual is computed; values it cannot map are flagged  
90 [CONVENTION REVIEW] with capped confidence rather than emitted as a silent wrong number.  
91 Crucially, the conversion factors were not guessed: they were derived with a guided physics-derivation  
92 assistant (GPD), which produced a reference that is *code-verified* against the repository’s plotting  
93 source and notebooks and *citation-audited* against the literature. Getting these right is a small but  
94 exacting physics task, and treating it as one — rather than as a prompting problem — is what made  
95 the registry trustworthy.

## 96 2.3 Code Generation, Integration, and Review

97 The LLM generates a `@staticmethod` for the relevant plotting class (a post-generation guard  
98 enforces the decorator); insertion uses Python’s AST module to locate the last method in the target  
99 class — never regex — preventing corruption. Notebook cells are inserted via `nbformat` and plots  
100 regenerated headlessly via `nbconvert`. Every update becomes a pull request, with low-confidence or  
101 unresolved-convention extractions flagged in the title. Crucially, we render a “highlighted” constraint  
102 plot in which all existing limits are greyed out and only the new proposed limit is shown in red  
103 (Figure 2), enabling rapid visual verification by domain experts without reading code.

## 104 2.4 Workflows and Cost

105 The **daily arXiv digest** queries, filters, classifies relevance, and processes matching papers. The  
106 **weekly preprint checker** scans existing data files for arXiv IDs and re-extracts on version updates to  
107 catch changed results; one PR detected that a JWST dark photon search [13] had changed a limit to a  
108 projection upon publication. The **historical backfill** searches INSPIRE-HEP [7] for older papers by  
109 citation count, splitting large jobs across daily runs.

110 The pipeline runs on a frontier model (Claude Opus [8]) to maximise extraction quality. With  $N=3$   
111 consensus and conditional vision, a paper triggers roughly five to nine model calls; text-only papers  
112 (the majority) avoid expensive image tokens. We do not present this as a cheap pipeline — per-paper  
113 cost is on the order of tens of cents to a few dollars, dominated by vision papers — but it remains far  
114 below the expert time it substitutes for, and text-first gating bounds the spend.

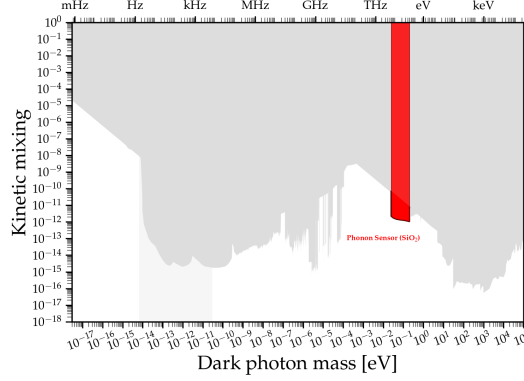


Figure 2: Highlighted dark photon constraint projection plot from a production pull request for a phonon-sensor proposal [14]. Existing limits in grey; new proposal in red. This format lets reviewers verify the proposal visually.

Table 1: Benchmark performance on 346 papers. Residual statistics are over the 271 comparable papers (243 with mass-range overlap); 0.3 dex  $\approx$  a factor of two.

| Aggregate metric       | Value     | By source / pooling       | Med. resid. | $\leq 0.3$ dex |
|------------------------|-----------|---------------------------|-------------|----------------|
| Coupling-type accuracy | 90.5%     | Text ( $N=148$ )          | 0.326 dex   | 49.7%          |
| Median residual        | 0.331 dex | Figure-vision ( $N=119$ ) | 0.338 dex   | 46.8%          |
| Within factor 2        | 48.4%     | Micro-avg (per paper)     | 0.331 dex   | 48.4%          |
| Within factor 3        | 61.7%     | Macro-avg (per type)      | 1.115 dex   | —              |
| Mean interp. coverage  | 76.2%     | AxionPhoton ( $N=127$ )   | 0.207 dex   | —              |
| Zero-overlap papers    | 10.3%     | AxionEDM ( $N=3$ )        | 4.649 dex   | —              |

### 3 Evaluation

**Benchmark and ground truth.** We evaluate on 346 papers spanning 14 coupling types, selected as all curated data files [5] with a known arXiv ID. The ground truth for each is the upstream repository’s own digitised curve — never data digitised by our own model — which avoids circularity but carries a consequence: the reference was itself digitised and rescaled from the same papers, so a *perfect* extraction still shows a nonzero residual equal to the upstream digitisation/convention gap. Our reported residuals are therefore an *upper bound* on true extraction error.

**Metric.** We compare in  $\log_{10}$ – $\log_{10}$  space: a log–log interpolation is built from the extracted points and evaluated at ground-truth masses, giving a per-point residual  $|\log_{10} g_{\text{ext}} - \log_{10} g_{\text{truth}}|$  (0.3 dex  $\approx$  factor of 2). We report median residual and *interpolation coverage* (fraction of GT masses the extracted curve spans) separately, because a curve can be accurate where it overlaps yet cover the wrong mass window. A reverse pass (GT interpolated onto the extracted masses) flags curves that run past the true mass range.

**Honest disposition taxonomy.** A curve is scored only against a same-coupling GT curve; we partition the rest so non-comparability never pollutes the residual statistics: *no comparable GT* (30 papers; predicted coupling absent, usually a misclassification), *convention mismatch* (10; same coupling, incompatible parameterisation), *point reference* (14; GT is a single operating-point limit, not a curve), *no extracted points* (16), and *no prediction* (3). Of 346 papers, 271 are comparable; the residual statistics below are over those.

**Classification labels.** Coupling type is scored against the repository’s directory structure. The property labels (*is new limit*, *is projection*, *data source*) are scored against an *independent* LLM labeler (a different model and prompt from the extractor it grades), making this a cross-model check rather than self-agreement; a 15-paper human audit found labeler–human agreement of 15/15, 15/15, and 14/15 on the three fields.

**Results.** Table 1 summarises the benchmark. Coupling-type classification is reliable (90.5%); 90% of comparable papers achieve mass-range overlap, with a median residual of 0.33 dex and 48% of

141 curves within a factor of two, at 76% mean coverage. The reverse pass agrees (median 0.34 dex),  
142 indicating the curves cover the right mass extent, not just the right values where they overlap.

143 The most consequential result is the *source breakdown*: figure-vision extraction (0.338 dex, 46.8%  
144 within factor two) is now on par with text extraction (0.326 dex, 49.7%). Reading dense log–log  
145 plots — historically the hard part of this task — is no longer the limiting factor; consensus voting and  
146 quality-tier selection over multiple reads close most of the figure-vision gap.

147 **Where the difficulty now lives.** The frontier has shifted to *rare coupling types*. The per-paper  
148 micro-average (0.33 dex) is dominated by the most common type (AxionPhoton, 0.21 dex,  $N=127$ );  
149 the per-type macro-average is 1.1 dex. Common types are strong (DarkPhoton 0.41, AxionElectron  
150 0.38), but small-sample, convention-heavy types lag: AxionNeutron 0.77, VectorBL 0.91, AxionMass  
151 0.94, and AxionEDM 4.6 dex. AxionEDM is the clearest case: the operator coupling  $g_{a\gamma\gamma}$  is flat in  
152 mass while the experimentally reported *response*  $\propto 1/m_a$  folds in the field amplitude — a convention  
153 our registry flags but does not yet auto-convert. These types also concentrate the coupling-type  
154 misclassifications, so they compound: a wrong type is scored against the wrong reference.

155 **Yardsticks.** Two floors bound interpretation. The run-to-run LLM *noise floor* (90th-percentile  
156 per-paper residual std over repeated extractions) is 0.32 dex — consensus voting is what keeps most  
157 papers inside it. The upstream *digitisation floor* (truly independent re-digitisation vs. repo, table/text  
158 sources) is only  $\sim 0.03$  dex, so the residuals we report are real extractor error, not a yardstick artefact,  
159 and the median sits essentially at the noise floor.

## 160 4 Lessons Learned

161 **Treat extraction as noisy and vote.** A single LLM read of a log–log plot is unreliable and irrepro-  
162 ducible. Reading each paper  $N=3$  times and reconciling by a medoid consensus with source-quality  
163 tiering was the single change that moved figure-vision from the dominant failure mode to parity with  
164 text. The medoid matters: averaging incompatible curves would manufacture a non-physical bound,  
165 whereas the medoid returns a real, central sample.

166 **The worst errors are deterministic physics, not perception — solve them with verified derivation.**  
167 The largest residuals were never about pixels. They were unit/convention mismatches — the axion-  
168 nucleon  $2m_N$  factor, the scalar  $d_e$  vs.  $\sqrt{\alpha}$  10–15 dex gap, dark-photon  $\varepsilon$  vs.  $\varepsilon^2$ , sentinel values  
169 mistaken for data — that are fixed, not stochastic, and depend on per-file repository conventions  
170 absent from the paper. The high-leverage fix was not a better prompt but a separate, auditable  
171 physics layer: we used a guided physics-derivation assistant (GPD) to produce a *code-verified*,  
172 *citation-audited* per-file convention registry, turning “unsafe guessing” about conversion factors into  
173 vetted closed-form maps, with a [CONVENTION REVIEW] flag for anything outside the verified set.  
174 Canonicalizing both sides before scoring also let us report *convention mismatch* as its own disposition  
175 (10 papers) instead of as inflated error. The general lesson for automated scientific extraction: it  
176 couples a *stochastic* perception problem (reading plots, best handled by sampling and voting) with a  
177 *deterministic* domain-knowledge problem (conventions, best handled by verified derivation), and the  
178 two demand different tools.

179 **Score honestly.** Grounding truth in the upstream repository (never in our own model’s digitisation)  
180 avoids circularity and makes the residual a defensible upper bound. Separating non-comparability  
181 (wrong-coupling, point-reference, convention-mismatch) from genuine error keeps the headline  
182 numbers meaningful, and scoring property labels against an independent LLM labeler rather than  
183 placeholders restores those metrics as a fair cross-model test.

184 **Engineering that paid off.** Text-first extraction with conditional vision keeps the majority of  
185 papers off the expensive vision path; AST-based insertion produces syntactically valid methods; and  
186 highlighted plots make review accessible to non-coders.

187 **What is still hard.** Rare coupling types remain the open problem: small samples, idiosyncratic  
188 conventions (the AxionEDM operator-vs-response case, scalar fifth-force files), and the misclassifica-  
189 tions that send them to the wrong reference. At 90% coupling accuracy roughly one proposal in ten  
190 still targets the wrong plot, so human-in-the-loop review is essential, not optional. A residual  $\sim 10\%$   
191 of papers also miss the mass window entirely (too few points or the wrong panel).

**Governance.** The pipeline is deployed and has generated limit proposals; none have merged, because no review process for AI-generated scientific data exists. We recommend a community editorial board analogous to PDG reviewers [15], where collaborations validate their own limits. More fundamentally, publishing machine-readable limit data alongside papers would bypass the extraction bottleneck altogether — the highest-leverage change the field could make.

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